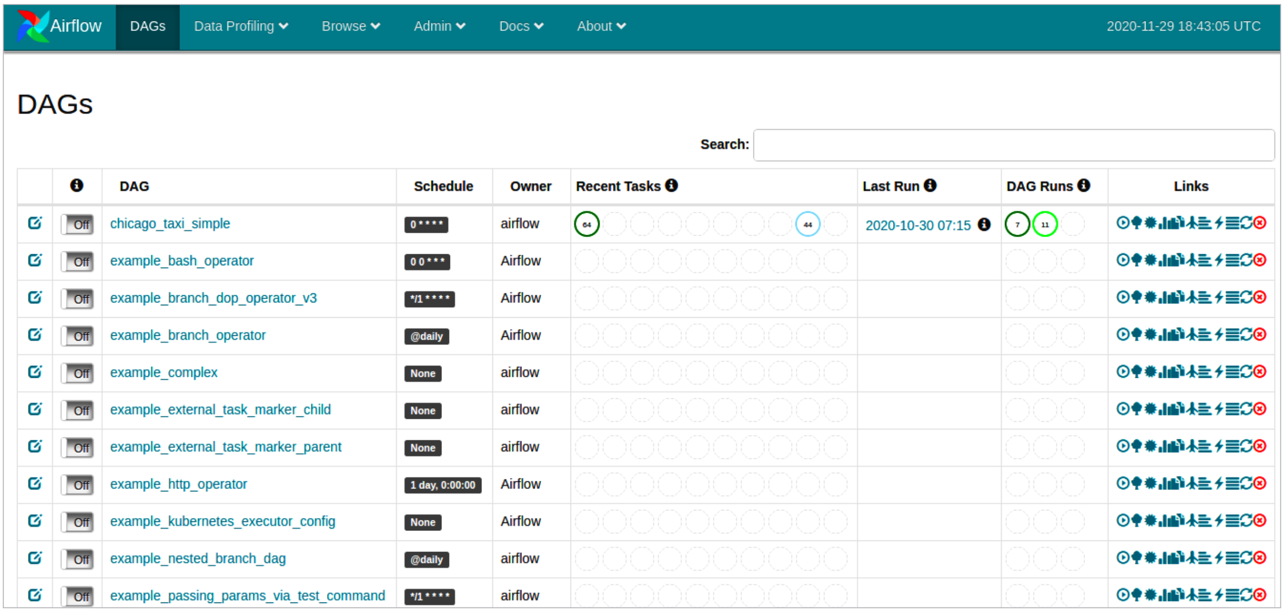

Figure 1: Apache Airflow dashboard

In this article, we try to develop a model that analyses taxi rides in the US city of Chicago, and then makes predictions whether a passenger will offer a tip to the driver or not. The model is then deployed using a MLOps pipeline. We use TFX and Apache Airflow for running the pipeline.

The system used for building the MLOps pipeline in this article is a Ubuntu 20.04.1 machine with Intel Core i5 processor and 4 GB RAM.

Installing TFX and Apache Airflow

Apache Airflow: Apache Airflow is an open source platform for managing workflows, developed by Airbnb. It allows us to create, schedule and monitor workflows for cron jobs, data fetching jobs, data processing tasks, and so on. Airflow can be used either with CLI or a Web based interface. The latter allows us to see the live status of workflows, in a fashion similar to Jenkins. Figure 1 shows the main dashboard of Apache Airflow.

Airflow workflows are written in the form of Directed Acyclic Graphs (DAGs) of tasks. In each DAG, a node represents a specific operation, which is performed using an operator. Other DevOps tools such as Jenkins can also integrate with Airflow using these operators. Thus, in a DAG, the first task could be running a Python script and the next one could be running a Bash script.

TensorFlow Extended: TensorFlow Extended (TFX) is an open source framework written in Python that allows us to build complete ML pipelines. A typical pipeline consists of the following stages.

- ***Data collection:*** Raw data is collected from its original source in this stage.
- ***Data analysis:*** Exploratory data analysis on the data received is done in this stage. Various aspects of the data such as data types, schema, distribution, etc, are studied.
- ***Data validation:*** This stage validates whether the data follows the distribution of previous data and matches the expected data schema.
- ***Data transformation:*** Here, we apply transformations for feature engineering to the data.
- ***Model training:*** The transformed data is fed to the appropriate ML algorithm so that it can learn the distribution of the data.
- ***Model evaluation and validation:*** In this stage, we first evaluate and then validate the performance of the model to check whether it matches the expected benchmarks. We cannot push the model and serve it if it does not do so.
- ***Pusher:*** The last step is to push the model to production so that other applications can use it to make predictions.

TFX allows us to define end-to-end ML pipelines that cover all the above steps. This helps, as it allows ML engineers to focus on actual model development rather than the other steps in the pipeline. It greatly automates the process of model deployment and hence allows us to perform continuous integration, continuous deployment and continuous training of newer models with ease. The TFX libraries that help us in achieving this are listed below:

- TensorFlow Data Validation
- TensorFlow Transform
- TensorFlow Model Analysis
- TensorFlow Serving
- Machine Learning Metadata

To install Apache Airflow and TFX,